Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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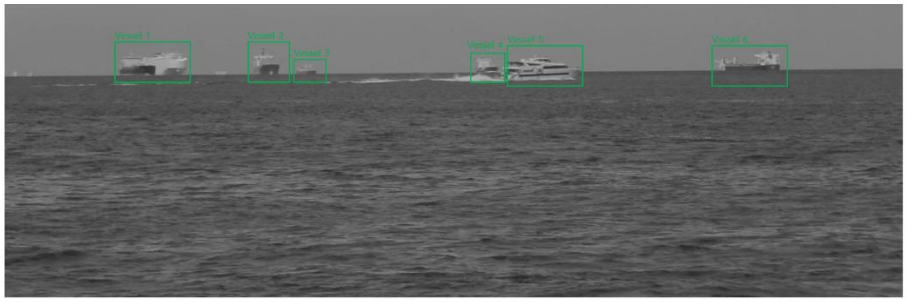
2. Technology Overview

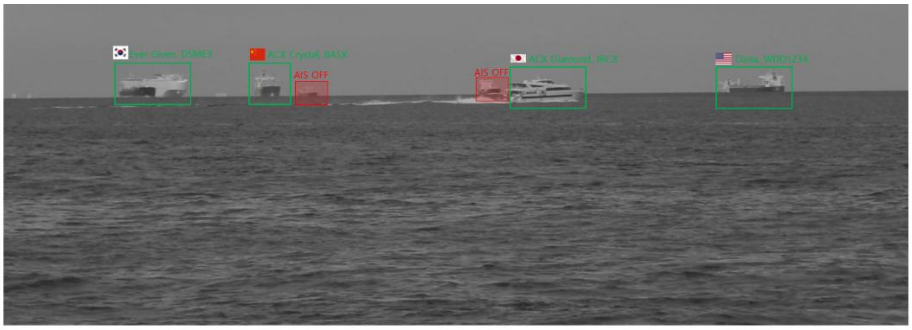
Title of Proposed Technology	**Development of an AI-Based Intelligent EOTS Maritime Surveillance System Integrating EO/IR and AIS**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Shipbuilding/ Marine Systems	*Shipbuilding/Marine System Software*	*Software*	*Software Solutions*
Concept of Proposed Technology	**Core Concept** Limitations of AI-Based EOTS and Directions for Technological Advancement • **Limitations of Existing EOTS and Directions for Technological Evolution** - Conventional EOTS systems are primarily limited to serving as observation equipment that provides EO/IR imagery, and the identification of maritime targets and situational awareness largely depend on the operator's experience and proficiency. - This technology aims to embed AI-based intelligent functions into EOTS, transforming it from a simple surveillance device into an intelligent maritime monitoring system capable of understanding operational situations autonomously. • **Concept of Transition to an Intelligent Surveillance System** - By utilizing AI-based image recognition and tracking technologies, EOTS can			

automatically detect and classify maritime targets and maintain consistent surveillance performance even during long-term operations, thereby establishing an advanced intelligent monitoring framework.

Current EOTS, standard models, and basic surveillance cameras



Surveillance cameras integrated with AI



AI-enabled surveillance cameras integrated with AIS

Key Functions and Potential Applications

EO/IR-Based Maritime Target Recognition and Tracking Technology

- **EO/IR Integrated Sensor-Based Surveillance Architecture**

  - By integrating daytime EO sensors with IR sensors capable of operating at night and in adverse weather conditions, an EO/IR-based surveillance architecture is implemented to enable continuous monitoring across diverse maritime environments.

- **AI Recognition Technology Specialized for Maritime Environments**

  - Approximately 80% of the training data is generated from 3D model-based synthetic data, while about 20% consists of real ship images and photographs used as supplementary data. This approach enables effective AI training and ensures stable target recognition and tracking performance even in environments with limited real-world vessel data.

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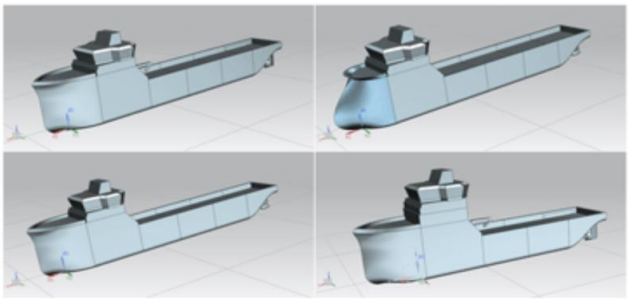

<Example of a 3D Model-Based Synthetic Data Generator>

Enhancement of Maritime Situational Awareness through AIS Integration

- **Concept of Intelligent Utilization of AIS Information**
  - AIS is an essential source of vessel information for maritime surveillance; however, in conventional systems, it is typically limited to simple data display. This technology utilizes AIS data as a core input for AI-based decision-making and performs integrated analysis with visual information.
- **Video-AIS Fusion-Based Situational Awareness**
  - By linking targets detected in EO/IR imagery with AIS information in real time, advanced situational awareness is achieved, enabling comprehensive understanding of maritime target locations, movement characteristics, and navigation patterns.

Architecture of an Automated and Intelligent Maritime Surveillance System

- **Automated Surveillance and Decision-Making Workflow**
  - The entire process, from EOTS image acquisition and AI-based target recognition and tracking to AIS-integrated analysis and situational assessment output, is fully automated to minimize operator intervention.
- **Scalable System Architecture**
  - The system is designed with a modular and scalable architecture, enabling application across various maritime surveillance platforms, including naval vessels, coastal monitoring stations, and port surveillance systems.

Rationale for Joint R&D Project

Critical Technical Barriers

- **Limitations of Conventional EOTS and Directions for Technological Advancement**
  - Traditional EOTS systems mainly function as observation tools that provide EO/IR imagery, while maritime target identification and situational awareness rely heavily on the operator's experience and skill.

  This technology seeks to integrate AI-driven intelligence into EOTS, shifting its role from basic surveillance equipment to an advanced maritime monitoring system capable of independently interpreting operational environments.
- **Limitations of Independent Development by a Single Company or Country**
  - AI-based intelligent maritime surveillance technology requires large-scale data training, advanced algorithms, and real-sea validation simultaneously, which creates structural

	limitations for independent development at the level of a single company or country. **Why Global Collaboration?** • **Global Trends in the Advancement of Maritime Surveillance Technologies** **-** In recent global maritime surveillance markets, the technological paradigm is shifting beyond simple sensor performance competition toward intelligent surveillance systems that incorporate AI-based automated situational awareness and decision-making technologies. Compared to the rapid pace of innovation in overseas maritime surveillance technologies, the development of core technologies in Korea remains relatively slow. • **Core Competencies of Overseas Partner Institutions** - Overseas research institutions, including Purdue University, possess fundamental AI technologies in areas such as maritime object recognition, multi-target tracking, sensor data fusion, and AI-based decision-making algorithms within the advanced manufacturing domain. - One of the core technologies of this project—automatic 3D shape generation and data augmentation—has been extensively developed and applied by Purdue University's AI Center and Manufacturing Center. These institutions possess strong technical capabilities and numerous practical applications for field deployment. Specifically, they have core technologies for acquiring meaningful data through real-time IoT edge computing from equipment signals and data obtained from actual manufacturing companies, as well as for processing and managing such data. They have also been providing AI modeling solutions tailored to industrial needs. By benchmarking these core technologies through this global R&D project, technological gaps can be rapidly narrowed, thereby enhancing project success rates and contributing to the creation of substantial and practical outcomes. • **Technology Integration through Collaborative Research** **-** The practical integration of EO/IR-based EOTS with AI and AIS convergence technologies is difficult to achieve without joint system design and iterative validation. Through international collaborative R&D, integrated performance can be verified in real maritime environments. • **Importance of Global Cooperation for Achieving TRL 7** **-** Through international collaborative R&D, the goal is to advance from laboratory-level technologies (TRL 4) to the stage of reliable validation in real maritime environments (TRL 7), which can be achieved through joint research with overseas partner institutions.
Key Words	1. **Electro-Optical Tracking System (EOTS)** 2. **EO/IR–AIS Data Fusion** 3. **AI-Based Intelligent Maritime Surveillance** 4. **Automated Situational Awareness** 5. **Scalable Maritime Surveillance System Platform**

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

The goal of this research and development project is to implement an AI-based EO/IR–AIS integrated intelligent EOTS maritime surveillance system that can automatically and intelligently recognize situations in maritime environments and quantitatively improve surveillance efficiency. To achieve this, clear quantitative performance targets will be established, and phased development will be carried out for each specific objective.

Detailed Objectives

- **Advancement of EO/IR-Based AI Maritime Target Recognition and Tracking Technology**
  - By integrating EO/IR imagery with AIS information, vessels will be automatically detected and classified, with target identification accuracy exceeding **95%** and multi-target tracking success rates of over **90%**.
  - To ensure stable recognition under long-range and low-resolution conditions, algorithms will be continuously refined with the goal of achieving an IR camera nighttime detection range of at least 5 km.
  - The developed recognition and tracking technologies will be applied to actual EOTS imagery, and performance will be validated based on real maritime environments.

- **Development of EO/IR–AIS Data Fusion-Based Automated Situational Awareness Technology**
  - A data fusion framework will be implemented by integrating and aligning target position and velocity information derived from EO/IR imagery with AIS vessel data in real time, targeting a video–AIS matching accuracy of over 90%.
  - Maritime situational awareness accuracy will be quantitatively improved through trajectory comparison and movement pattern analysis.
  - An automated surveillance system capable of stable situational awareness without operator intervention will be established.

- **Securing AI-Based Intelligent Surveillance Decision-Making and Automation Performance**
  - AI algorithms will be used to analyze vessel movement patterns and navigation characteristics, and intelligent surveillance decision logic will be developed with a target of over 95% overall judgment accuracy and a false alarm rate below 5%.
  - Automation functions will be advanced to maintain consistent decision outcomes even in multi-target and long-term monitoring environments.

- The system will be designed to automatically present surveillance results in a format that operators can easily understand.

- **System Integration and Technology Readiness Level (TRL) Achievement through Real-Sea Demonstration**

  - The developed EO/IR–AIS integrated intelligent surveillance technologies will be applied to real maritime environments for phased demonstration tests, and system stability will be verified through long-duration continuous operation trials.
  - Compliance with key performance indicators will be confirmed through demonstrations in naval vessel and coastal surveillance environments.
  - By the completion of the project, a Technology Readiness Level of TRL 7 will be achieved.

R&D Target Specifications

In this project, key performance indicators (KPIs) will be established to objectively evaluate the technical maturity of the AI-based EO/IR–AIS integrated intelligent EOTS maritime surveillance system. For each indicator, target performance levels, weighting factors, and measurement and validation methods will be clearly defined.

Evaluation criteria	Target Value	Score Weight	Measurement and Validation Method
Maritime Target Detection Accuracy (EO/IR-Based)	95%↑	20%	Analysis of detection success rates compared to ground truth data during real-sea trials
Multi-Target Tracking Success Rate	90%↑	15%	Statistical analysis of track maintenance rates and tracking loss rates during simultaneous monitoring of multiple vessels
EO/IR–AIS Matching Accuracy	90%↑	20%	Analysis of spatial and temporal alignment accuracy between image-based tracks and AIS tracks
Intelligent Situational Assessment Accuracy	95%↑	20%	Evaluation of decision accuracy based on predefined surveillance scenarios
False Alarm Rate	5%↓	10%	Statistical analysis of false alarm occurrence rates during long-term continuous operation tests
Nighttime Detection Range (MWIR-Based)	5km↑	10%	Measurement of maximum recognition distance (DRI-R: Recognition) through nighttime maritime field trials
System Stability and Continuous Operation Capability	24 hour operation	5%	Verification of system error occurrence through long-duration real-sea continuous operation tests

R&D Plan

Research management and strategies

- Maritime Operational Environment-Based Data Utilization Approach: Training data will be constructed primarily using synthetic data generated from 3D vessel models (approximately 80%), supplemented by EO/IR imagery and AIS data (approximately 20%) collected during actual operations.
- Phased Performance Enhancement and Iterative Validation Approach: Detection and tracking, data fusion, and intelligent decision-making functions will be developed in stages,

and performance evaluation results from each phase will be iteratively reflected in subsequent design to achieve continuous performance improvement.

- System Integration-Oriented Demonstration Approach: Rather than focusing solely on algorithm-level development, research will be conducted with system-level integration with EOTS hardware in mind, aiming to achieve a level of system maturity suitable for real-sea demonstrations.
- Global Collaborative Research-Based Technology Integration Approach: Core AI technologies from overseas partner institutions will be organically combined with the EOTS development and field demonstration capabilities of domestic SMEs, enabling the simultaneous achievement of technological maturity and reliability through international collaborative R&D.

Major Research Activities and Milestones by Year

Year	Objectives	Major Tasks	Milestone
Year 1	EO/IR-Based AI Recognition and Tracking Technology	· Development of a synthetic data generation and training environment based on 3D vessel modeling · Development of baseline AI models for vessel detection and multi-target tracking using integrated EO/IR imagery and AIS data · Algorithm integration testing based on EOTS video input	- Achievement of vessel detection accuracy of over 90% - Securing baseline performance for multi-vessel tracking - Verification of AI–EOTS integration technology based on 3D synthetic data
Year 2	Advancement of EO/IR–AIS Fusion and Intelligent Situational Awareness	· Development of real-time alignment and fusion logic for EO/IR imagery and AIS data · Implementation of algorithms for analyzing maritime movement characteristics and navigation patterns · Enhancement of intelligent situational awareness and automated decision-making functions	- Achievement of EO/IR–AIS matching accuracy of over 85% - Implementation of automated situational awareness functions - Improvement of intelligent surveillance decision accuracy
Year 3	Real-Sea Demonstration and Completion of the Intelligent EOTS System	· Establishment of an integrated EO/IR–AIS fusion-based intelligent EOTS system · Conducting real-sea demonstration tests in naval vessel and coastal surveillance environments · Verification of stability during long-duration continuous operation	- Achievement of overall decision accuracy of over 95% - Securing a false alarm rate of 5% or lower - Achievement of TRL 7 level technology maturity

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	[Roles and Responsibilities of Overseas Research Institution – Purdue University] • **Development of Core AI Algorithms for Vessel Recognition, Classification, and Tracking** - Provide synthetic data generation and AI training technologies based on 3D vessel models, and design and advance a training strategy centered on synthetic data. - Develop vessel recognition, classification, and tracking algorithms that integrate EO/IR imagery and AIS information, and deliver core AI technologies applicable to EOTS systems. • **Research on EO/IR–AIS Data Fusion and Intelligent Decision-Making Logic** - Design algorithms for aligning and fusing EO/IR-based vessel information with AIS trajectory data to enhance automatic vessel classification and surveillance decision accuracy. - Establish a technical foundation for automated surveillance functions by developing intelligent decision-making logic based on vessel navigation characteristics. • **Research on Maritime Multi-Agent AI Based on LLM-Embedded AI Agents** - Develop AI agents equipped with Ask Gates and Policy Gates to build intelligent and interactive language models. - Provide specialized multi-agent AI technologies trained in the maritime domain that serve as intermediaries in data pipelines between vessel data and classification algorithms.
Expected R&R of Korean SME Partner	[Roles and Responsibilities of Korean SMEs] • **Integration of AI into EO/IR-Based EOTS Systems and Advancement of Intelligent Capabilities** - AI-based vessel recognition and classification algorithms will be integrated into existing EOTS systems to upgrade observation-oriented platforms into intelligent maritime surveillance systems. A system architecture integrating EO/IR image processing, AIS linkage, and AI-based decision outputs will be designed to ensure stable operation in real operational environments. • **System Integration, Real-Sea Validation, and Securing Operational Suitability** - Intelligent surveillance systems with integrated AI functions will be deployed within existing EOTS operational environments, and key performance indicators (KPIs), including vessel recognition accuracy, tracking performance, and automated decision accuracy, will be validated through real-sea trials. System reliability, stability, and operational suitability will be continuously improved by incorporating test results from various operational scenarios. • **Commercialization and Expansion of Applications Based on System Advancement** - The advanced AI-based EOTS system developed through R&D will be structured as a software-oriented solution, and phased deployment and upgrade commercialization will be pursued for existing EOTS customers. The application scope of intelligent EOTS systems will be expanded to domestic and global maritime surveillance markets, and continuous technology diffusion will be promoted through follow-up R&D and business collaborations.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** • **Securing Core Technologies for AI-Based EO/IR–AIS Integrated Intelligent EOTS** - Vessel recognition, classification, and tracking performance will be achieved through an AI training strategy that combines 3D vessel model-based synthetic data (80%) with real EO/IR imagery and photographic data (20%). Through this approach, conventional observation-oriented EOTS systems will be upgraded into intelligent vessel surveillance systems. • **Achievement of TRL 7-Level Technology Maturity through Real-Sea Validation** - Compliance with key performance indicators (KPIs) will be verified through real-sea trials, achieving a technology maturity level suitable for practical use. System stability and reliability will be secured, enabling immediate commercialization. • **Establishment of Joint IP and Commercialization Foundations** - Core technologies developed through collaborative research will be secured as joint intellectual property (IP), forming the basis for global market entry and subsequent business expansion. **Commercialization and Economic Outcomes** • **Commercialization of High-Value Intelligent EOTS Products** - AI, EO/IR, and AIS convergence technologies will be applied to existing EOTS product lines to commercialize premium intelligent maritime surveillance systems. - Product unit prices and profitability will be enhanced through functional and performance upgrades. • **Commercialization Potential and Phased Market Entry** - Based on TRL 7-level demonstration results at project completion, immediate commercialization will be possible without the need for additional large-scale development. - Initial markets will be secured through existing EOTS customers, followed by global market entry through joint commercialization with overseas partner institutions. **Industrial and Social Impact** • **Improvement of Operational Efficiency through Maritime Surveillance Automation** - The application of intelligent vessel surveillance systems will improve labor-dependent monitoring structures and enhance surveillance efficiency. - Consistent surveillance quality will be maintained even in long-term and multi-vessel monitoring environments. • **Enhancement of National Maritime Surveillance Capability and Safety** - Advanced coastal, port, and maritime monitoring capabilities will contribute to strengthening maritime safety.The diffusion of intelligent maritime surveillance technologies will reduce social costs and enhance public safety.

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